

VINEETH KIRANDUMKARA

IOT PRODUCT DEVELOPMENT ENGINEER

CONTACT

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EDUCATION

- 2024 - 2025
UNIVERSITY COLLEGE LONDON
THE BARTLETT FACULTY OF THE BUILT ENVIRONMENT
 - MSc Connected Environments
- 2014 - 2019
VIRGINIA TECH
BRADLEY DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING
 - Bachelor of Computer Engineering

SKILLS

- Embedded Systems Design
- IoT Architectures and Protocols
- Product Development and Prototyping
- System Optimization
- Hardware/Software Integration
- User Interface Design
- Data Analysis

LANGUAGES

- C++
- C
- Python
- Javascript

PROFILE SUMMARY

Embedded systems engineer specializing in Connected Environments and IoT, with extensive experience in firmware, hardware/software interfacing, RTOS, and large-scale product development. Skilled in C/C++, Python, and system design, with a record of driving reliability, scalability, and innovation in smart products. Passionate about merging robust embedded software with user-centric device and system design to deliver impactful real-world solutions.

WORK EXPERIENCE

- Grenova** Nov 2023 - Aug 2024
Embedded Software Engineer Richmond, VA
 - Co-developed firmware for TipNovus 2.0, leading architecture for dependency management and cross-repo integration.
 - Implemented advanced memory pool structures to improve system flexibility and storage efficiency.
 - Authored command/reaction protocols and system diagnostics for pipette automation, driving product reliability.
 - Developed secure JSON storage for system info and low-level drivers (sensor, sdmmc, eeprom).
- Iontra Inc** Nov 2022 - Nov 2023
Embedded Software Engineer Denver, CO
 - Automated the firmware verification processes across 2000+ new boards; transitioned systems to manufacturing.
 - Developed firmware for real-time monitoring of pouch cell metrics, safety limits, and bug diagnostics.
 - Modularized CMake project architecture, developed RTOS-based task scheduling, and added OS/HAL abstraction improvements.
- Platform Aerospace** Feb 2021 - Sept 2023
Embedded Systems Engineer Hollywood, MD
 - Empowered world record-breaking UAV endurance via complete codebase rewrite and advanced memory/resource management.
 - Delivered 50% reduction in dynamic memory and 40% faster processing cycle by leveraging C++ pattern optimizations.
 - Designed embedded web interface for device orchestration, led Hardware-in-the-Loop sim initiatives, organized code review, and embedded agile workflow across team.

- Designed and implemented a data reporting tool (Python + Alteryx) to streamline validation workflows and improve cross-team visibility into large-scale test results.
- Proposed and led the transition from Waterfall to Agile workflows, boosting collaboration and efficiency across local and international teams.
- Developed and rolled out a Skill Matrix framework to assess competencies, identify gaps, and structure targeted training for continuous team development.
- Collaborated with Product Validation Engineering to create reporting methods for tested ECUs, supporting more informed product decision-making.

- Partnered with stakeholders to design and deliver a native iOS app with AWS backend integration, enabling vehicle browsing through dynamic filtering.
- Collaborated with a cross-functional team to build an intuitive, user-centered UI and enhance overall app usability.
- Implemented a secure base-32 encrypted login system for store employees, unlocking advanced features such as test drive management.
- Coordinated iOS-backend development through REST API refactoring and rapid iteration cycles, including 24-hour bug resolution to meet critical demo deadlines.

PROJECTS

Turbulent Spaces (MSc Dissertation)

- Investigated wind flow and turbulence in urban canyons using low-cost ultrasonic anemometers.
- Designed experiments in a wind tunnel with various obstacles to mimic real-world street scenarios.
- Applied statistical analysis to quantify environmental effects of buildings and trees on airflow.
- Demonstrated use of scalable IoT sensor networks for real-time urban microclimate monitoring.

Gesture Recognizer

- Built an embedded machine learning system using flex sensors and Edge Impulse to capture hand gestures.
- Developed and deployed the full hardware and data pipeline on Arduino Nano 33 BLE.
- Enabled real-time recognition for user interaction and prototyping of wearable IoT interfaces.

Hot Stone

- Developed an IoT-based device to remotely share warmth between two users.
- Engineered sensing and actuation for tactile feedback, targeting long-distance relationships.
- Integrated hardware, firmware, and user interface for synchronized remote interactions.

Traveler

- Created a mobile app using Flutter/Dart with Supabase backend for contextual location reminders.
- Incorporated Google Maps and Places APIs for real-time geolocation and notifications.
- Focused on intuitive user experience and scalable product architecture for travel convenience.

Lumos

- Engineered an interactive lighting control system with magnetometers and rotary controls.
- Utilized MQTT protocol for real-time communication and LED control across a distributed system.
- Designed smooth user interface logic for mapping gestures to color changes in a connected environment.